Superstore Sales & Customer Analysis

This project analyzes sales data to identify revenue trends, high-value customers, and profit performance. The goal is to uncover business insights that can help improve sales strategy and decision-making.

Dataset : Global superstore sales dataset
Tools used : SQL | PostgreSQL

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Business Problem Statement

📌 Problem :

The business generates strong sales but struggles to clearly understand what drives profitability across products, discounts, and shipping costs.

🎯 Goal of the Analysis:

Identify key factors affecting revenue, profit, and operational efficiency to support better business decisions.

🔍 Key Business Questions Explored:

- Which product categories generate the highest profit?
- Which customers contribute the most revenue?
- How do shipping costs impact profit margins?

✨ *And additional insights derived from 12+ SQL queries*

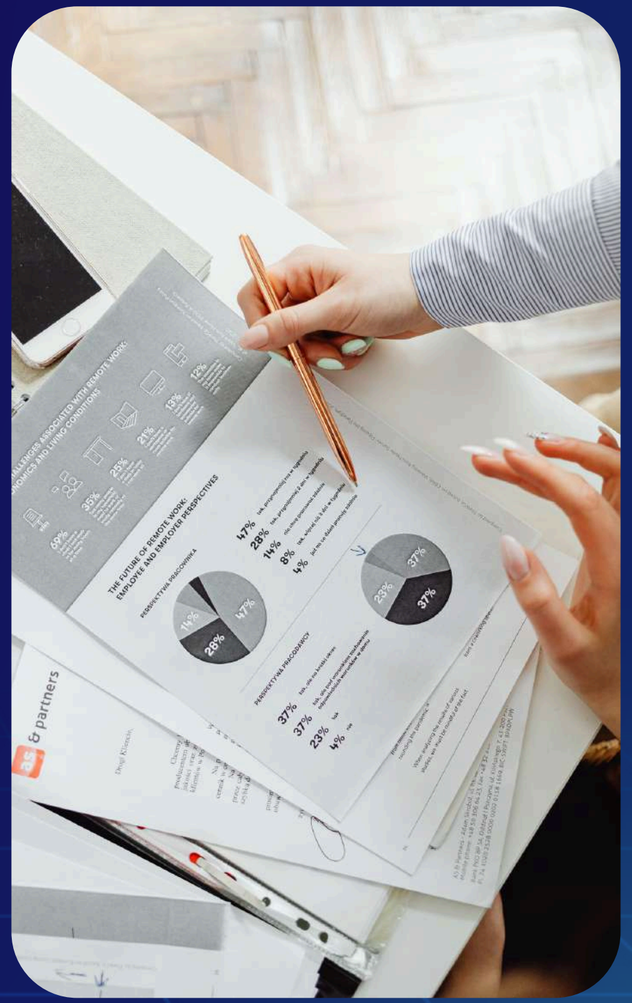
Dataset Overview :

Customers : 4,873

Total Orders : 25,035

Total Revenue : $61.8M

Period : 2011 -2014



Month-Over-Month Profit Growth

key insights :

• Profit peaked in August ,indicating the strongest business performance during this month

• Profit dropped into negative in January and July ,indicating that costs excessed revenue during these months

Business Recommendation :

- Launch promotional campaigns in December to reduce the potential drop in January sales.
- Introduce special mid-year offer in June to boost demand before the july slowdown
- Optimize operational costs during January and July to reduce the risk of negative profit margins

Customer Revenue Distribution

key insights :

- The top 10% customers (Decile 1) generate 37% of total revenue, indicating strong revenue concentration among a small group of customers

- This indicates a Pareto-like distribution (80/20 rule) where a small group drives most of the revenue.

Business Recommendations :

- Introduce loyalty programs, exclusive offers, or premium support to retain top-value customers.
- Focus on Decile 3–5 customers because they have potential to move into higher-value segments.
- Avoid spending excessive marketing budget on low-value customers with minimal revenue impact.

Customer Segmentation Analysis (RFM)

key insights :

A large proportion of customers fall into the “Needs Attention” segment, indicating an opportunity for the business to improve customer retention strategies and increase repeat purchases.

customer_segment text	customers bigint	percentage text
Needs Attention	1842	41.43%
Champion	866	19.48%
At Risk	541	12.17%
Lost Customer	503	11.31%
New Customers	354	7.96%
Loyal Customers	340	7.65%

Business Recommendation :

- Reward and retain Champion customers with loyalty programs and exclusive offers.
- Re-engage “Needs Attention” customers through targeted promotions.
- Run win-back campaigns for At Risk and Lost customers.

Profit margin by each category

key insights :

- Technology generates the highest revenue and profit, with the strongest profit margin (2.88%).
- Furniture shows the lowest margin (1.42%) despite decent revenue, indicating potential cost/pricing issues.

category character varying (100)	total_revenue numeric	total_profit numeric	profit_margin_pct numeric
Technology	23047940	663778.73318	2.88
Office Supplies	18686322	518473.8343	2.77
Furniture	20088638	285204.7238	1.42

Business Recommendation :

- Focus promotional campaigns on high-margin categories to maximize profit.
- Review pricing and cost structure of Furniture products to improve profitability.

Discount vs Profit Impact

key insights :

- Low or no discount transactions contribute the majority of profits.
- Medium and high discounts significantly reduce profitability.

discount_level text	total_orders bigint	total_revenue numeric	total_profit numeric	avg_discount_pct numeric
No Discount	15211	33741831	1770695.2732	0.00
Low Discount	7084	18684517	511444.10478	14.83
Medium Discou...	2842	5244660	-187271.06182	36.96
High Discount	4021	4151892	-627411.02488	58.78

Business Recommendation :

- Avoid excessive discounting as it leads to profit loss.
- Maintain discount levels below ~20% to protect margins.

Shipping Cost Impact on Profit

key insights :

- Higher shipping costs often reduce profit margins.
- Same Day shipping has the highest cost in Technology and Office Supplies.
- Faster delivery does not always lead to higher profitability.

category character varying (100)	ship_mode character varying (50)	total_ship_cost numeric	avg_ship_cost numeric	avg_profit numeric	rank_by_avg_profit bigint
Furniture	Same Day	34718.90	68.34	37.00	1
Furniture	First Class	102861.69	71.88	32.01	2
Furniture	Standard Class	202784.21	33.97	29.80	3
Furniture	Second Class	99954.68	50.79	21.71	4
Office Supplies	Second Class	98122.04	15.55	17.73	1
Office Supplies	Same Day	33946.00	20.27	16.92	2
Office Supplies	Standard Class	181068.80	9.69	16.44	3
Office Supplies	First Class	92311.50	20.08	15.44	4
Technology	Second Class	116035.07	57.16	67.98	1
Technology	Standard Class	230774.65	37.73	66.28	2
Technology	First Class	112929.35	76.51	61.84	3
Technology	Same Day	47308.81	91.33	56.06	4

Business Recommendation :

- Limit the use of high-cost shipping options such as Same Day delivery
- Promote cost-efficient delivery options like Standard or Second Class

Conclusion & Key Takeaways :

Key Findings :

💸 High discounts significantly reduce profit margins.
👥 A small number of customers contribute a large share of total revenue.
🚚 Shipping costs and delivery choices impact overall profitability.
📉 Some products generate revenue but result in financial losses.

Business Recommendations :

🎯 Use targeted discounts instead of heavy blanket promotions.
⭐ Retain high-value customers with loyalty programs and exclusive offers.
📦 Optimize shipping strategies to reduce logistics costs.
🔍 Review pricing or discontinue consistently loss-making products.